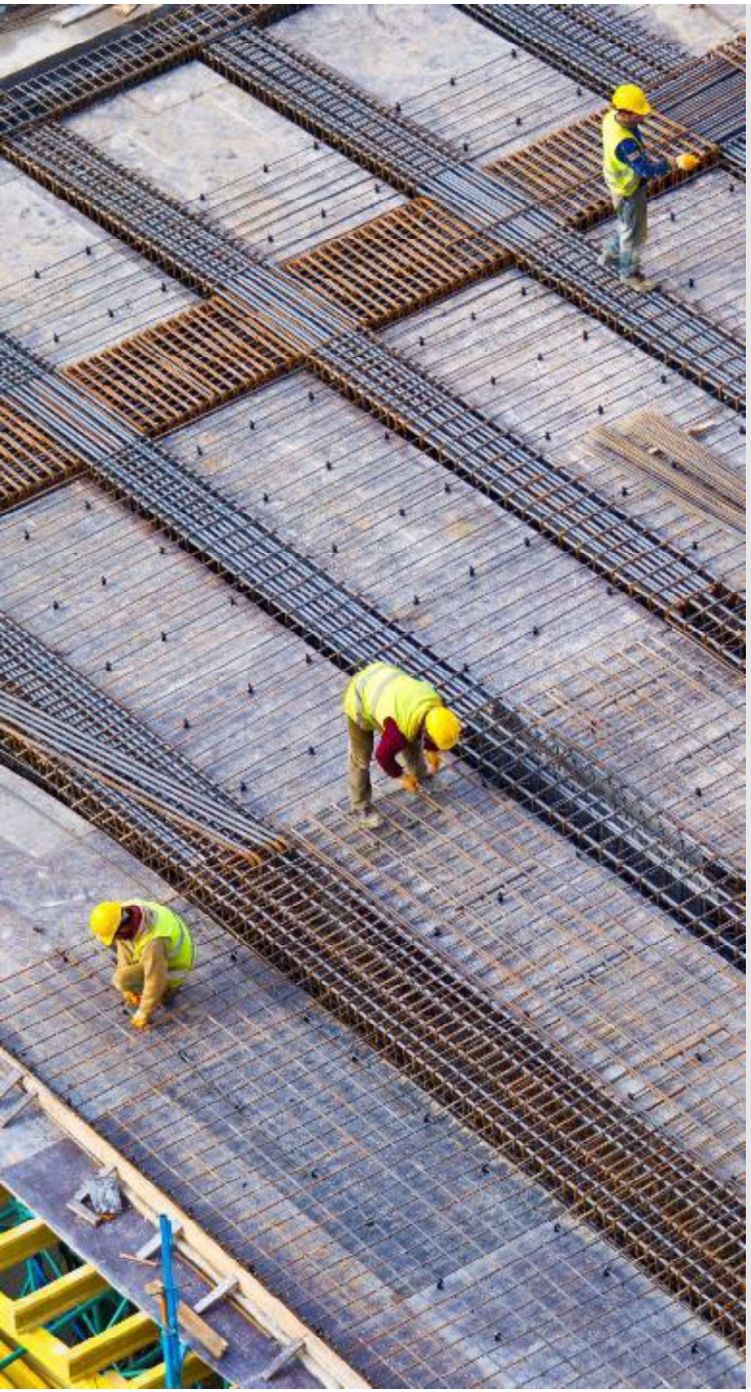
tngz

Digital Worksite

Market



Construction Industry is Ripe for Digitilization

Majority of construction companies are in early stages of their DX journey. Manual data collection and poor connectivity on worksites causes errors and under utilization.

An average construction site with 500 machines losts **$13.480.000 every year*** because of wrong work-hour reporting & idling.

Profitability for contractor companies has dropped below 4%**. Completing projects on time and on budget, optimization through digitalization became a priority for construction industry.

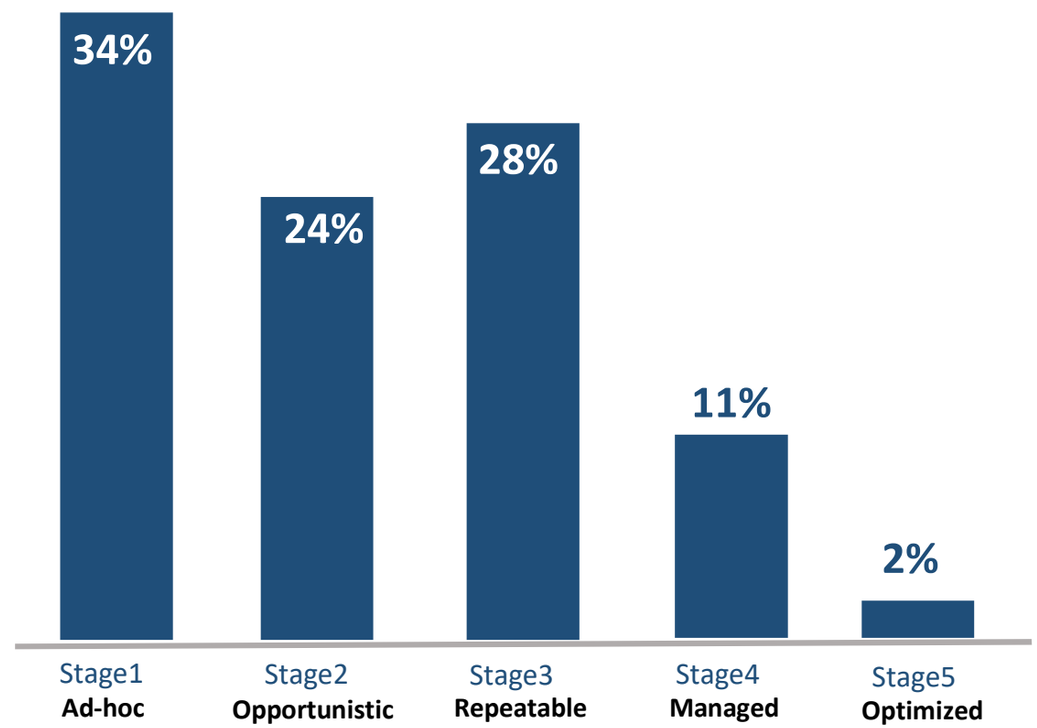
**Equipment watch 2020*
***Construction Financial Management Association, 2020*

Construction Industry is Keeping Pace in Digitalization

"Construction tech startups are poised to shake up a $1.3-trillion-dollar industry."
TechCrunch 2020

"investors including Goldman Sachs Group Inc. are pouring capital into construction startups, betting that the industry is on the verge of sweeping changes." – Wall Street Journal 2019

Almost 60% of construction companies worldwide are only starting their DX journeys.

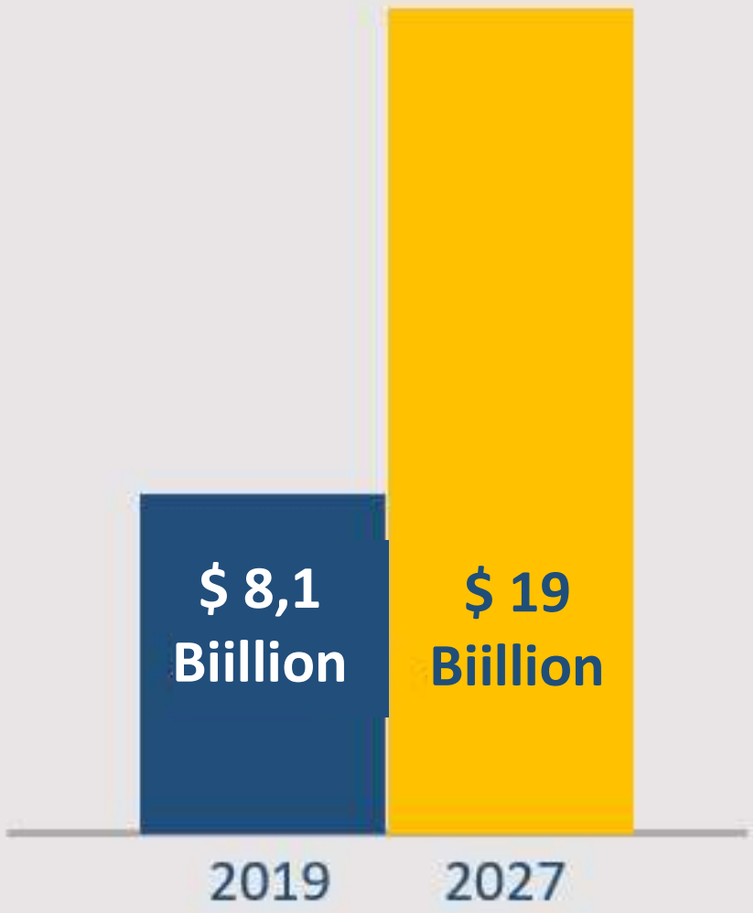


"Digital Transformation: The Future of Connected Construction" – IDC 2020

IoT in Construction Market

The global IoT in construction market size is expected to reach $19 billion in 2027

CAGR **16.5%**



"IoT in Construction Market" – Allied Market Research 2019

Benefits of Digital Work-Hour Tracking

- Operating expenses of an average construction machine is $120K per year.
  By eliminating errors in work-hour reports and reduced idling, **$38K per year for each machine can be saved.**

- Better asset management helps **reduce equipment rentals upto 20%**.

- Integration with different IoT solutions and 3rd part applications improve smart decisions on worksites.

- Real time work hour tracking can help to optimize maintenance processes.

- Low idling helps reduce **carbon emissions upto 15%** in construction machines.

Cost of Idling Equipment

Subtype	Size Class	Hourly Idling Rate	Hours Idle per day	Annual Savings
4-Wd Articulated Wheel Loaders	175 – 199 HP	$57.90	1	$15,285.60
Tractor-Loader-Backhoes	16’ to Under 17’	$35.63	2	$18,812.64
Crawler Mounted Hydraulic Excavators	28.1 – 33.0 MTons	$96.72	1.5	$38,301.12
Straight Mast Rough Terrain Lift Trucks	10,000 – 12,999 Lbs.	$34.43	2.5	$22,723.80
Wheel Tractors	75 to 124 HP	$16.70	1.5	$6,613.20
On-Highway Light Duty Trucks	100 – 199 HP	$9.90	1	$2,613.60

“Equipment Watch: Controlling the cost of Idling Equipment” - 2020

Product

Innovative Telematics
& Equipment Management Platform

- Tingz is a cloud based, device agnostic equipment management platform for construction companies, mines and equipment rental companies.
- Customers can track their equipments, reduce idling and generate error-free run hour reports.
- System Integrator Interface helps SI's to manage multiple customer sites and provide additional services to their customers.
- Tingz enables orchestration of different IoT devices, applications and ERP solutions

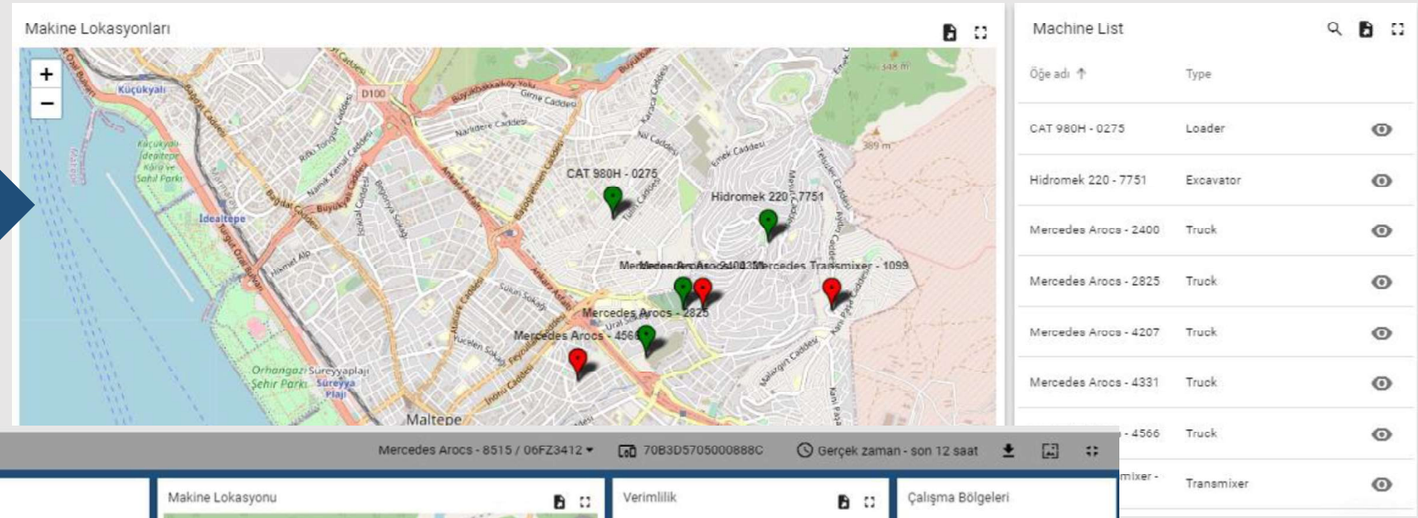
Hardware & Algorithms

Tingz uses GSM independent, easily mounted, specially configured LoRaWAN sensors for data collection.

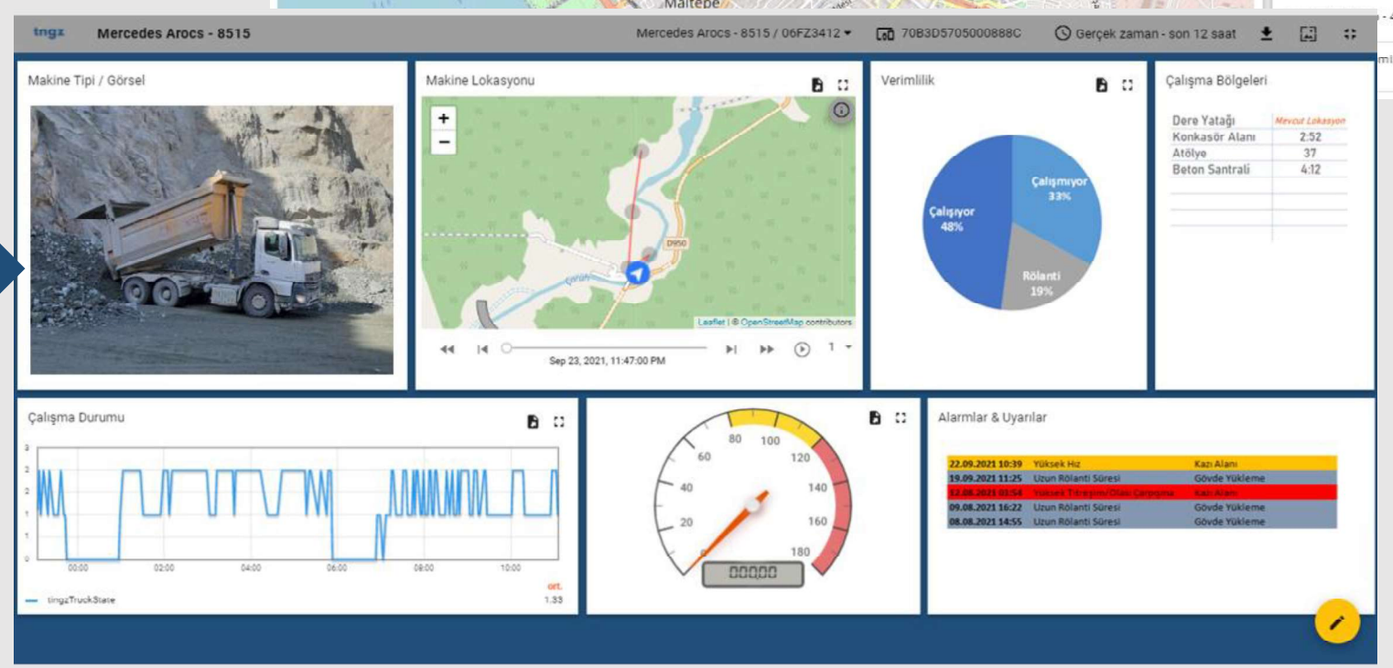
Data from construction equipment is analysed with Tingz algorithms to identify working conditions, idle times and accidents. Collected data will also be used for machine learning and predictive maintenance.

User Interface

Top-down view of worksite

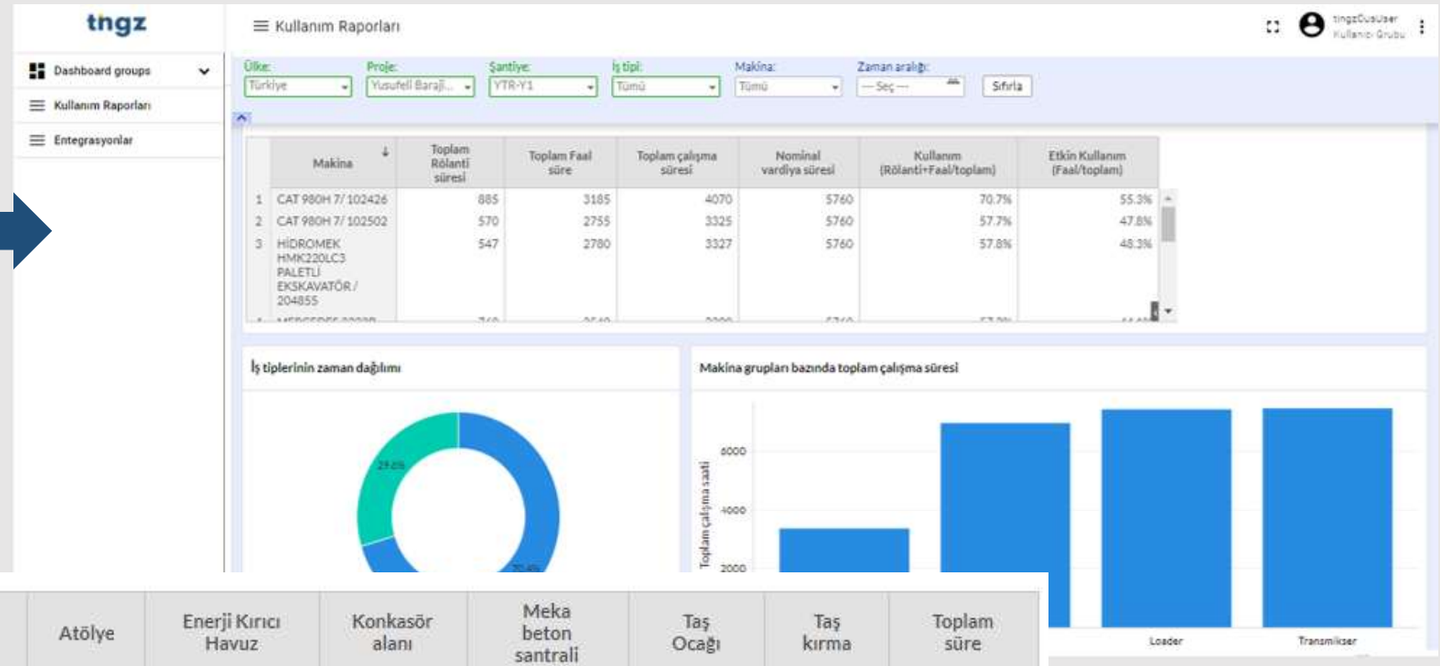


Detailed analysis of each equipment



User Interface

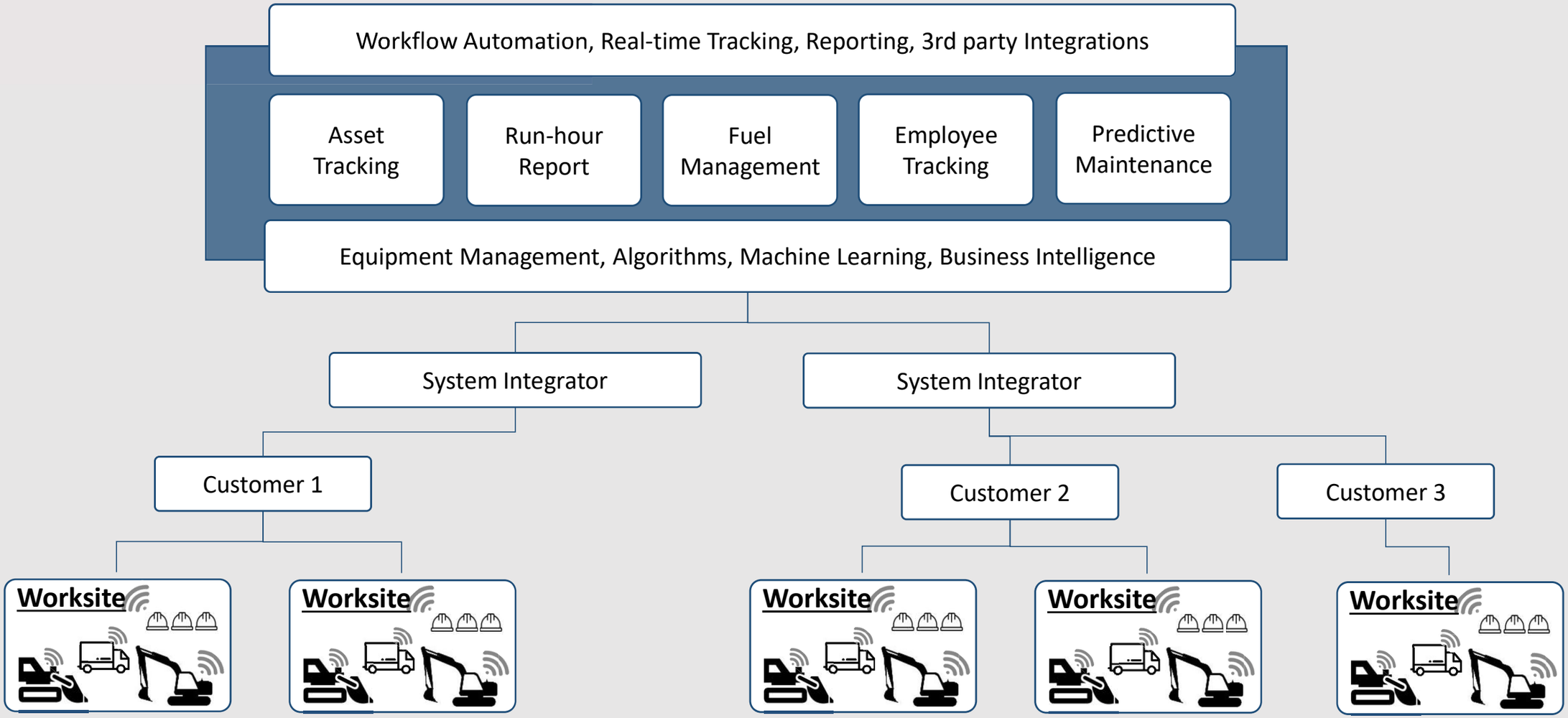
Productivity and comparison analysis based on work zones, equipment groups and projects.



Run-hour reports with day&night shifts.

Ekipman	Tarih	Vardiya	Durum	715	Asfalt santrali	Atölye	Enerji Kırıcı Havuz	Konkasör alanı	Meka beton santrali	Taş Ocağı	Taş kırma	Toplam süre
CAT 980H 7/ 102502	02 Mar 2021	01-Sabah	Faal							405		405
			Rölanti							15		15
		01-Sabah								420		420
		02-Akşam	Faal							415		415
			Rölanti							15		15
		02-Akşam								430		430
	02 Mar 2021									850		850
	03 Mar 2021	01-Sabah	Faal					180				180
			Rölanti							20		20
		01-Sabah						180		20		200
		02-Akşam	Faal					60		350		410
			Rölanti					45		100		145

System Architecture



tngz Product Road Map

- Asset Tracking
- Run-Hour Reports
- Fuel Management
- Employee Tracking
- Predictive Maintenance & Machine Learning
- Integration with 3rd party IoT solutions and applications

Go-to-market Strategy

TINGZ is Getting Traction in Turkey

Product development is completed in partnership with Limak Holding. Which is one of the biggest construction companies in Turkey. Limak Holding has decided to use Tingz in its worksites. Deployment starts in November.

Starting Point: Turkey

Turkey has one of the biggest construction markets in the World

$30 billion revenue in 2020.

Turkish construction companies amongs top contractors in the World

46 Turkish companies listed in top 250 of the World biggest construction companies.

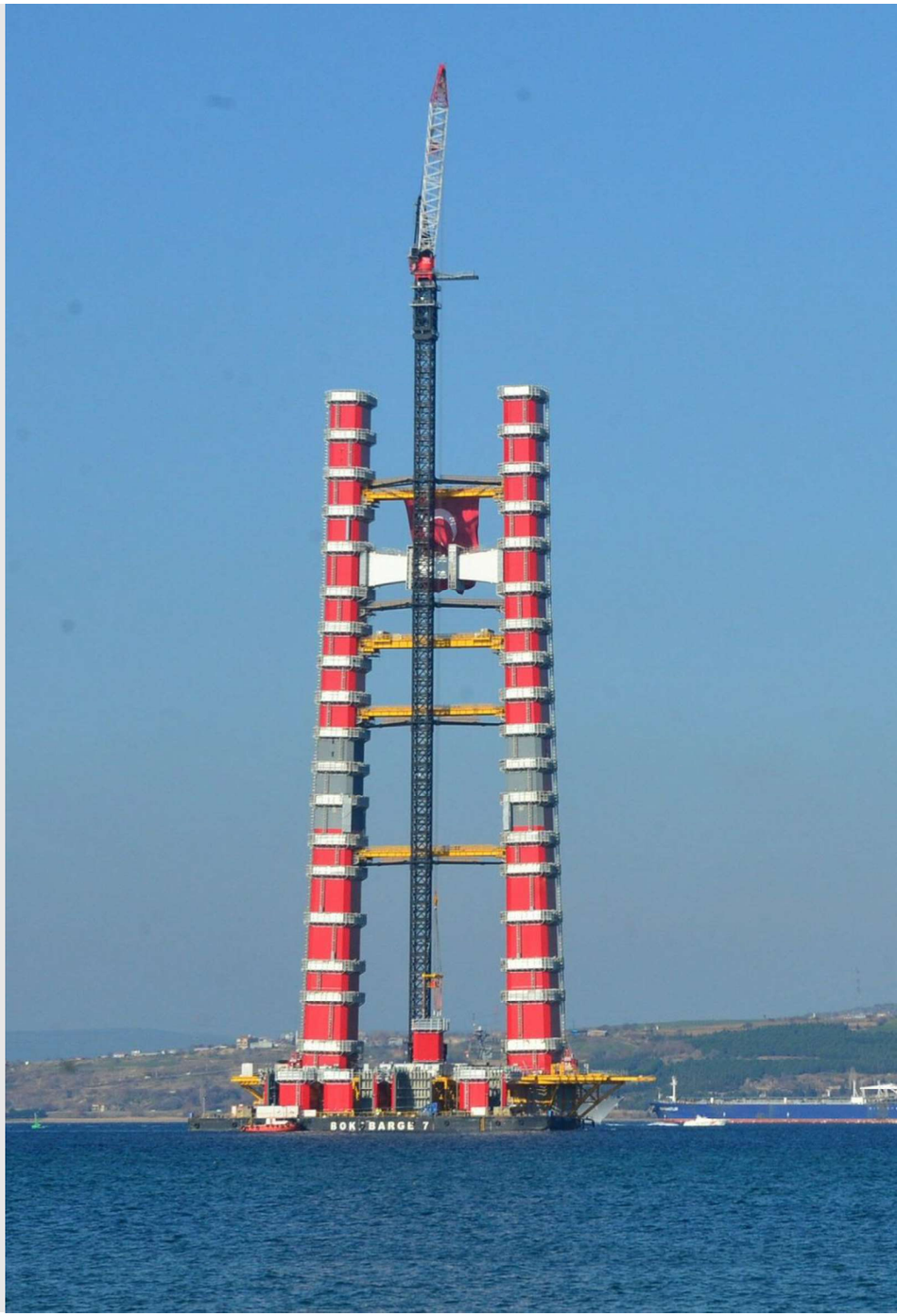
International Reach

8.830 projects in 100 countries.

Active in different segments

Energy, infrastructure, transportation and residence.

Working with Turkish companies presents an opportunity to scale up and reach global markets.



Run-Hour Tracking Application Business Model

License : Annual subscription **$240/sensor** (79% Gross Margin)

After Sales Support: 10% of annual subscription fee – provided by SIs

IoT Hardware: 130$ -provided by manufacturer

Example use case: Worksite with 500 construction machines:

In most cases 350 out of 500 machines are required to be tracked and monitored.
In this case, project costs will be:

350 IoT Sensors and one time set up cost: $ 48.000
Annual license fee: $350 x 240: $84.000
After sales support: $8.400

Total Contract: ***$140.400***
350 machines x 38.530 USD = 13.480.000 USD yearly savings

Avg. Revenues According to Project Types

Dam Construction, ≈840 sensors ***$201.600 /year***

Airport Construction, ≈600 sensors: ***$144.000 /year***

Highway Construction, ≈250 sensors: ***$60.000 /year***

Road Map

January 2021

- Limak Holding Parnership
- Prototype
- Developig Algorithms
- Field Research

June 2021

- Tests & Validation
- 150 Connected Devices

March 2022

- Expanding in Limak + New Customers
- Roll Out: SI Interface

June 2022

- Personnel Tracking & Fuel Management
- Expanding in SI network. 2.500 Connected Devices

December 2022

Revenue Forecast

- 2021: 150 connected machines, $18K/year – (Limak Holding)
- 2022: 150 → 2.500 connected machines, : $300K/year - (Limak+ 2 new customers)
- 2023: 2500 → 10.000 connected machines : $2.4M /year – (Expanded SI network and new applications)
- 2024/25:→ 45.000 connected machines: $10 M

Competition

Competitive Advantages

- Experienced team in Industry 4.0 and cloud technologies
- Unlimited access to worksites and equipments with Limak partnership.
- Specially developed algorithms for machine working conditions.
- 800 man.hour field experience. 2.400 hours testing.

Competition - Turkey

GSM Operators

- ↑ Access to local market
- ↓ GSM depended connectivity. Doesn’t comply with worksite processes

- ↓ GSM depended connectivity. Doesn’t comply with worksite processes
- ↓ High costs

Competition - Global

MACHINEMAX

«Acquired by SHELL in June 2021»

- ⬆ Good hardware & software solution for worksites
- ⬇ Limited with only equipment management

WORLD SENSING

- ⬆ Strong and extensive hardware portfolio
- ⬇ Software solution doesn’t comply with worksite processes

Team

tingz

Çağatay Kaan Karataş
CEO

4 years R&D, 10 years B2B busines development experience. B.Sc.in Mechanical Engineering. M.Sc in Engineering Management

Ersin Gülaçtı
CTO

20 years experience in product development and cloud technologies. B.Sc. in Computer Science, MBA degree

Haluk Çavuşoğlu
Business Development / Partner Management

Entrepreneur, sales coach, CRM Turkey Borad Member. B.Sc.in Mechanical Engineering, MBA degree

Advisors

Alper Yeğin

VP of Advanced Technology Development in Actility; Vice-Chair of BoD, Chair of Technical Committee LoRa Alliance

Metin Taşkın

Airties, Co-Founder and CTO

Assoc. Prof. Dr. İlhan Yılmaz

Advisor at Start-up Laws and International investments

Ömer Erkmen

Mentor, Investor